3.4 Complex Numbers

Name _____ Period ____ Date _____

Algebra II 3.4 · Student sheet

I can…

- define and use the imaginary unit i.
- add, subtract, and multiply complex numbers.
- divide complex numbers and find complex solutions of quadratic equations.

Vocabulary · fill in the missing word

Imaginary unit i — $i =$ _____, so $i^2 = -1$

Complex number — $a + bi$, where a and b are _____; a is the real part and b is the imaginary part

Imaginary number — $a + bi$ with $b \neq 0$; _____ imaginary if $a = 0$

Complex conjugates — $a + bi$ and $a - bi$; their _____ is $a^2 + b^2$, a real number

Part 1 · Powers of i

i^1	i^2	i^3	i^4	i^5	i^6	i^7	i^8

Pattern: _____

$i^{23} =$ _____ $i^{40} =$ _____ $i^{102} =$ _____

Part 2 · Notes and examples (fill in each note, then work the example)

The imaginary unit: $i =$ _____, $i^2 =$ _____

Example 1

a. Find $\sqrt{-25}$ and $\sqrt{-18}$. b. Solve $x^2 + 16 = 0$.

Complex numbers: A complex number has the form _____.

Example 2

Simplify.

a. $(7 + 3i) - (2 - 5i)$ b. $(-4 + i) + (6 - 2i)$

Example 3

Multiply $(2 + 3i)(4 - i)$.

Complex conjugates: The conjugate of $a + bi$ is _____. To divide, multiply by the _____ of the denominator.

Example 4

Divide $\frac{5 + i}{2 - i}$.

Part 3 · Partner practice

Partner 1

Multiply $(1 - i)(3 + 2i)$.

Partner 2

Find i^{30}.